

Course Schedule

Date	Topic	Suggested Readings	Assignments	Term Project
Week 1	Course Introduction (In-Person Session @NC 1003)			(Project Kick-off List!)
Week 2	<u>Lecture 1 – Word Vectors</u> ↓ <u>Video #1</u> ⇨, <u>Video #2</u> ⇨, <u>Video #3</u> ⇨	1. <u>Efficient Estimation of Word Representations in Vector Space</u> ⇨ (original word2vec paper) 2. <u>Distributed Representations of Words and Phrases and their Compositionality</u> ⇨ (negative sampling paper)		
Week 3	<u>Lecture 2 – Word Vectors (cont'd), and Word Senses</u> ↓ <u>Video #1</u> ⇨, <u>Video #2</u> ⇨	1. <u>GloVe: Global Vectors for Word Representation</u> ⇨ (original GloVe paper) 2. <u>Improving Distributional Similarity with Lessons Learned from Word Embeddings</u> ⇨ 3. <u>Evaluation methods for unsupervised word embeddings</u> ⇨	#1 Assigned on 9/4	
Week 4	<u>Lecture 3 – Neural Networks</u> ↓ <u>Video #1</u> ⇨, <u>Video #2</u> ⇨, <u>Video #3</u> ⇨	1. <u>Matrix Calculus Notes</u> ↓ 2. <u>Differential calculus Notes</u> ↓ 3. <u>Backprop Notes</u> ⇨ 4. <u>Derivatives, Backpropagation, and Vectorization Notes</u> ↓ 5. <u>Network architectures</u> ⇨ ⇨ 6. <u>Learning Representations by Backpropagating Errors</u> ⇨ (seminal Rumelhart et al. backpropagation paper)		

Week 5	Lecture 4 – Dependency Parsing ↓ Video #1 ⇨, Video #2 ⇨	1. Incrementality in Deterministic Dependency Parsing ⇨ 2. A Fast and Accurate Dependency Parser using Neural Networks ⇨ 3. Globally Normalized Transition-Based Neural Networks ⇨ 4. Universal Stanford Dependencies: A cross-linguistic typology ⇨ 5. Universal Dependencies website ⇨ 6. Jurafsky & Martin Chapter 14 ⇨	#1 Due & #2 Assigned on 9/18	
Week 6	Lecture 5 – Language Models and Recurrent Neural Networks ↓ Video #1 ⇨, Video #2 ⇨	1. N-gram Language Models ⇨ (textbook chapter) 2. The Unreasonable Effectiveness of Recurrent Neural Networks ⇨ (blog post overview) 3. Sequence Modeling: Recurrent and Recursive Neural Nets ⇨ (Sections 10.1 and 10.2) 4. On Chomsky and the Two Cultures of Statistical Learning ⇨ 5. Sequence Modeling: Recurrent and Recursive Neural Nets ⇨ (Sections 10.3, 10.5, 10.7-10.12) 6. Learning long-term dependencies with gradient descent is difficult ⇨ (one of the original vanishing gradient papers) 7. On the difficulty of training Recurrent Neural Networks ⇨ (proof of vanishing gradient problem) 8. Vanishing Gradients Jupyter Notebook ⇨ (demo for feedforward networks) 9. Understanding LSTM Networks ⇨ (blog post overview)		Proposal Due (9/25) <u>Proposal Preparation Instructions</u>

Week 7	Lecture 6 – Recurrent Neural Networks (cont'd). ↓ Video #1 ⇨, Video #2 ⇨		#2 Due & #3 Assigned on 10/2	
Week 8	Lecture 7 – Neural Machine Translation, Sequence-to-Sequence Models, and Attention ↓ Video #1 ⇨, Video #2 ⇨, Video #3 ⇨	1. Statistical Machine Translation ⇨ (lectures 2/3/4) 2. Statistical Machine Translation ⇨ (book by Philipp Koehn) 3. BLEU ⇨ (original paper) 4. Sequence to Sequence Learning with Neural Networks ⇨ (original seq2seq NMT paper) 5. Sequence Transduction with Recurrent Neural Networks ⇨ (early seq2seq speech recognition paper) 6. Neural Machine Translation by Jointly Learning to Align and Translate ⇨ (original seq2seq+attention paper) 7. Attention and Augmented Recurrent Neural Networks ⇨ (blog post overview) 8. Massive Exploration of Neural Machine Translation Architectures ⇨ (practical advice for hyperparameter choices) 9. Achieving Open Vocabulary Neural Machine Translation with Hybrid Word-Character Models ⇨ 10. Revisiting Character-Based Neural Machine Translation with Capacity and Compression ⇨		
Week 9	Lecture 8 – Self-Attention and Transformers ↓	1. Attention Is All You Need ⇨ 2. The Illustrated Transformer ⇨ 3. Transformer (Google AI blog post) ⇨		

	Video #1 ➞	4. Layer Normalization ➞ 5. Image Transformer ➞ 6. Music Transformer: Generating music with long-term structure ➞		
Week 10	Lecture 8 - Self Attention and Transformers (cont'd) Video #2 ➞		#3 Due & #4 Assigned on 10/23	
Week 11	Lecture 9 - Pretraining ↓ Video #1 ➞	1. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding ➞ 2. Contextual Word Representations: A Contextual Introduction ➞ 3. The Illustrated BERT, ELMo, and co. ➞ 4. Martin & Jurafsky Chapter on Transfer Learning ➞		
Week 12	Lecture 9 - Pretraining (cont'd) Video #2 ➞			
Week 13	Lecture 10 - Prompting and RLHF ↓ Video #1 ➞, Video #2 ➞	1. Language Models are Few-Shot Learners ➞ 2. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models ➞ 3. Finetuned Language Models Are Zero-Shot Learners ➞ 4. Learning to summarize from human feedback ➞	#4 Due & #5 Assigned on 11/13	
Week 14	Lecture 10 - Prompting and RLHF (cont'd) Video #3 ➞			

Week
15

Fall Break

Week
16

Lecture 11 – Natural
Language Generation



Video #1 ⇄, Video #2



1. The Curious Case of Neural Text Degeneration ⇄
2. Get To The Point: Summarization with Pointer-Generator Networks ⇄
3. Hierarchical Neural Story Generation ⇄
4. How NOT To Evaluate Your Dialogue System ⇄

#5 Due
on 12/4

**Final Report &
Presentation
Due 12/4**

Presentation
Preparation
Instructions

Report
Preparation
Instructions

Finals
Week

**Final Exam
In-Person
Time and Location
TBA**